# Adrian Selk

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## Education

**B.Sc. Mathematics (second degree)** Oct 2026 - present
*Humboldt-Universität zu Berlin, Germany*

- Coursework on calculus and linear algebra

**B.Sc. Business Information Systems - Data Science** Oct 2023 - Sep 2026
*Baden-Württemberg Cooperative State University (DHBW) Stuttgart, Germany*

- Cooperative (dual) degree programme run jointly with IBM; 210 ECTS; final grade 1.6 (1.0 = best)
- Relevant coursework: machine learning, mathematics (calculus, linear algebra, statistics, operations research), algorithms and data structures, big data, distributed systems, software engineering
- Thesis: *Augmenting Vision-Language Models for Wildfire Burn Assessment: Capabilities and Limits.* Conducted at IBM Research Zurich, supervised by Dr. Mattia Rigotti. Grade 1.5.

**Abitur (university entrance qualification)** 2022
*Marie-Curie-Gymnasium Ludwigsfelde, Germany*

- Grade 1.4; advanced courses in Mathematics and English

## Publications

[1] C. Lohse, **A. Selk**, A. Ba, J. Wahl, M. Ruffini. *Licence to Scale: A Microservice Simulation Environment for Benchmarking Agentic AI.* Workshop on Scaling Environments for Agents (SEA), NeurIPS 2025.
Contribution: designed and built the simulation environment the study runs on, during a research internship at IBM Research Dublin; released as open source (see below).

## Open-Source Software

- **L2SBench** (now *ITBench Exploratory Simulator*): microservice simulation environment accompanying the NeurIPS 2025 SEA workshop paper; github.com/itbench-hub/ITBench-Exploratory-Simulator
- **ONNX-MLIR**: upstream contributor to the open-source AI compiler (1,000+ GitHub stars); operator optimisation in C; github.com/onnx/onnx-mlir

## Research Experience

**Research Intern | Adapting VLMs to Earth-Observation Signals** Aug - Sep 2026
*IBM Research Zurich, Switzerland (remote)*

- Taught a small generalist VLM to read physically motivated renderings of Earth-observation data: spectral indices, SAR composites, temporal differences

**Bachelor's Thesis | Vision-Language Models for Earth Observation** Feb - May 2026
*IBM Research Zurich, Switzerland (remote)*

- Built a multi-stage system that combines vision-language models with conventional classifiers for assessing wildfire damage from satellite imagery, along with a benchmark to evaluate it
- Central finding: the model's final assessment matched its initial visual read in 99.4% of cases, replicated across two VLMs; the later processing stages almost never revised it (visual anchoring effect)

**Research Intern | ONNX-MLIR Compiler** Aug - Nov 2025
*IBM Research Tokyo, Japan*

- Optimised several operators in C for the open-source ONNX-MLIR AI compiler; changes merged upstream

**Research Intern | Agentic AI** May - Aug 2025
*IBM Research Dublin, Ireland*

- Designed and built a causality-driven microservice simulation environment for benchmarking reinforcement-learning and LLM-agent systems on cloud autoscaling under service-level objectives
- Co-authored the resulting paper, accepted at the NeurIPS 2025 SEA workshop; environment released as open source (*L2SBench*)

**Research Intern | Climate Risk Analytics** Nov 2024 - Feb 2025
*IBM Research Zurich, Switzerland (remote)*

- Built a workflow to ingest raw climate data and compute climate indices supporting risk assessment for critical infrastructure (roads, railways, bridges); analysed and visualised long-term climate trends

## Industry Experience

**Intern | Client Engineering** Jun - Sep 2024
*IBM, Böblingen, Germany*

- Developed a text-to-SQL agent translating natural-language questions into queries over a large product database; supported client pilot projects

**Intern | Customer Success Management** Jan - Mar 2024
*IBM, Ehningen, Germany*

- Co-developed a LangChain tutorial and built early prototype applications on top of large language models

**Intern | Marketing & CRM Analytics** Oct 2022 - Jun 2023
*TD Reply, Berlin, Germany (pre-university)*

## Technical Skills

- **Programming:** Python (advanced), Java, SQL, R, JavaScript
- **Machine learning:** PyTorch, scikit-learn, LangChain, LangGraph, Gymnasium (RL environments), prompting and evaluation of large multimodal models
- **Tools:** Git, Docker, Jupyter, LaTeX, Kubernetes (basic)

## Languages

German (native) • English (IELTS 8.0, May 2026) • Japanese (beginner) • French (basic)

## Awards

- 2nd place, Mesh Stuttgart Hackathon (Apr 2024): end-to-end retrieval-augmented chatbot over machine manuals for first-line technical support